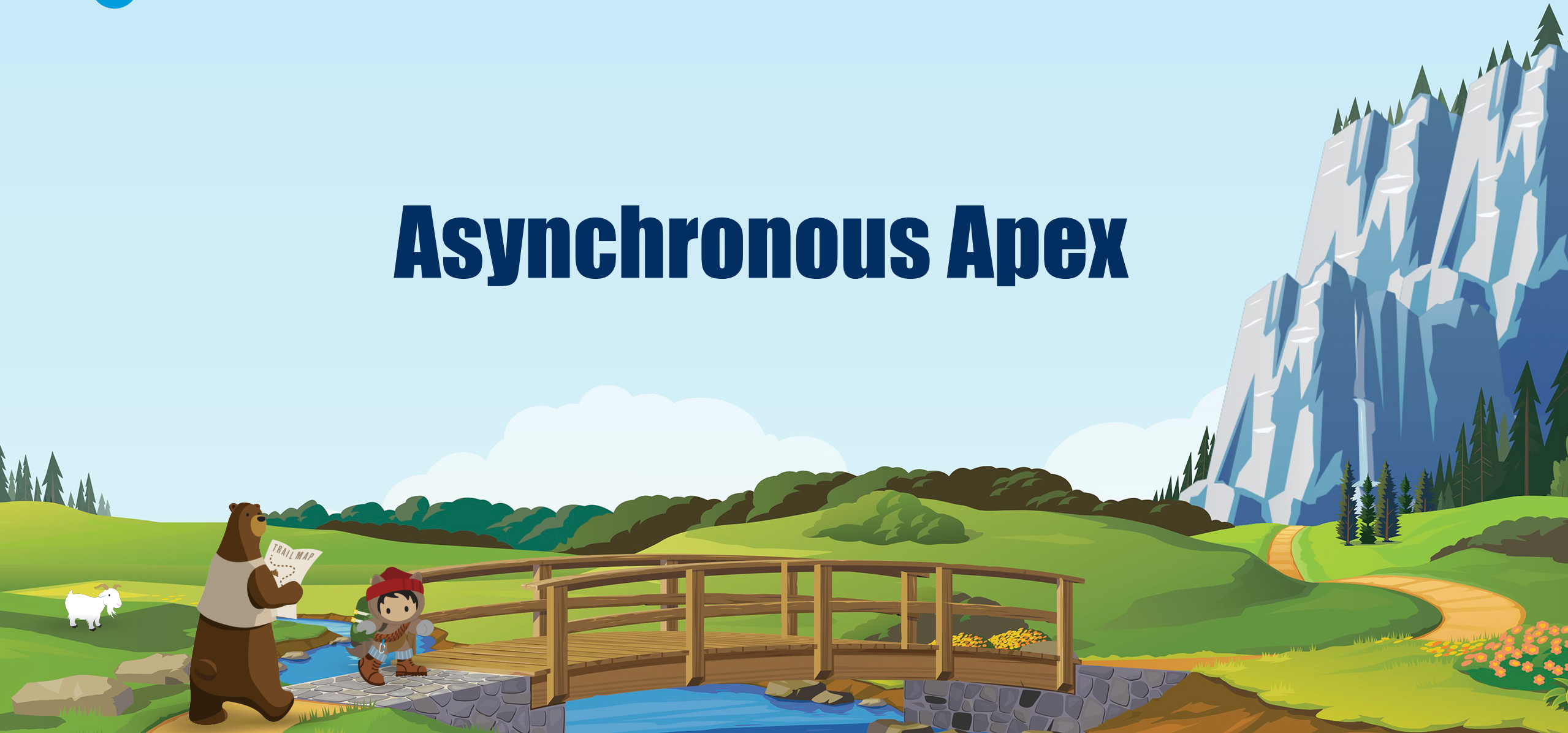


Asynchronous Apex



Agenda



- What is Asynchronous Apex
- Why do we need Async Apex
- Types of Async Apex



What is Asynchronous Apex :



Asynchronous Apex is a way of running Apex code in the background, in a separate thread from the user's request, allowing long-running, resource-intensive, or delayed operations to execute without blocking the user or hitting synchronous governor limits.

- Executes in a separate transaction
- Avoids synchronous governor limits
- Handles long-running or heavy operations
- Allows callouts from triggers
- Improves user experience and performance
- Supports scalable background processing
- Enables delayed or scheduled execution



Why do we need Asynchronous Apex :



Asynchronous Apex is required when certain operations are too heavy, too slow, or not allowed to run in a normal synchronous transaction. It helps offload complex logic to the background, ensures smooth user experience, and allows Salesforce to process large or time-consuming tasks without hitting governor limits.

Key Reasons

- Handle long-running operations
- Manage large data volumes
- Avoid synchronous governor limits
- Perform callouts from triggers
- Resolve mixed DML restrictions
- Improve overall user performance
- Run scheduled or delayed jobs
- Support complex multi-step processing



Types of Asynchronous Apex :



Salesforce provides multiple asynchronous mechanisms to run background operations based on the nature, size, and timing of the task. These options help handle heavy processing, large data volumes, scheduled jobs, and third party callouts.

Following are the various async Apex Options we have in salesforce :

- **Future Methods** - Run lightweight operations in the background, ideal for callouts and actions that cannot be performed in the main transaction.
- **Queueable Apex** - Provides flexible, chainable background processing with support for complex data structures and multi-step workflows.
- **Scheduled Apex** - Executes Apex automatically at a specified time or frequency using CRON expressions, perfect for recurring jobs.
- **Batch Apex** - Handles massive data volumes by processing records in manageable chunks, enabling scalable and controlled background processing.



thank you

